

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	14 September 2026
Team ID	SWUID2026136950
Project Name	Automated Network Request Management in ServiceNow
Maximum Marks	4 Marks

Technical Architecture:

Architecture of the Automated Network Request Management application, showing the User Layer (local), the ServiceNow Cloud Platform (Presentation, Application Logic and Data layers) and the External Interfaces.

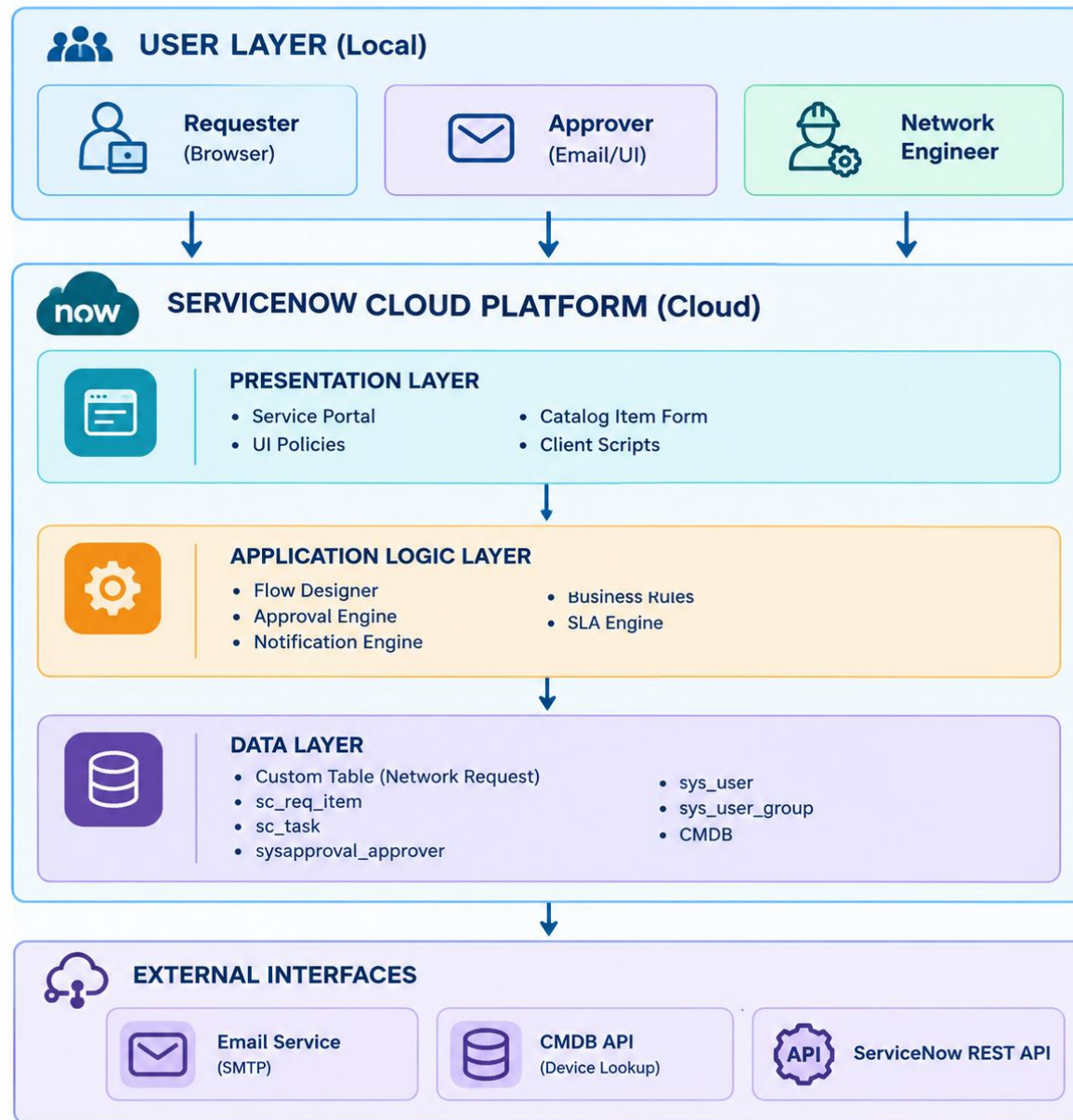


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Web portal through which requesters submit network requests and approvers/engineers act on them	Service Portal (HTML5, CSS3, JavaScript, AngularJS)
2.	Application Logic-1	Client-side form behaviour and validation	ServiceNow Client Scripts, UI Policies
3.	Application Logic-2	Server-side business rules and default values	ServiceNow Business Rules (JavaScript)
4.	Application Logic-3	End-to-end workflow automation (approval, task creation, notification)	ServiceNow Flow Designer
5.	Database	Data type and configuration of the Network Request custom table and related tables	ServiceNow Cloud Database (MySQL-based)
6.	Cloud Database	Managed database service	ServiceNow Cloud Infrastructure
7.	File Storage	Attachment storage for request documents	ServiceNow Attachment Storage

8.	External API-1	Device lookup from the CMDB	ServiceNow CMDB API
9.	External API-2	Email delivery for notifications and approvals	ServiceNow Email Engine (SMTP)
10.	Machine Learning Model	Not applicable in this phase	N/A
11.	Infrastructure (Server / Cloud)	Application deployment on cloud	ServiceNow Cloud (SaaS)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Open-source frameworks and platform APIs used	AngularJS – Service Portal UI framework Bootstrap – responsive UI components jQuery – DOM manipulation and AJAX calls GlideRecord API – server-side database access GlideSystem API – server-side utility functions
2.	Security Implementations	Security and access controls applied	Role-Based Access Control – requester, approver, engineer ACLs – read, create, write restricted by role Group-level permissions – Network Requesters, Network Approvers, Network Team Record-level security on the custom table Audit logging of approval actions and state changes Secure email delivery – SMTP over TLS Session management – ServiceNow built-in Admin override for emergency changes; data encryption at rest and in transit
3.	Scalable Architecture	3-tier scoped application on a	Architecture – Presentation, Application, Data tiers ServiceNow SaaS (multi-tenant)

S.No	Characteristics	Description	Technology
		multi-tenant cloud platform	Horizontal and database scaling handled by the platform Flow Designer supports concurrent executions Supports 100+ requests per day Modular scoped application with independent deployment
4.	Availability	Platform-level availability and recovery	Uptime SLA – 99.9% (ServiceNow Cloud) Multi-region data centers and platform load balancing Backup and disaster recovery by ServiceNow No downtime for active user sessions; reliable SMTP delivery with retry logic Built-in SLA engine for tracking
5.	Performance	Design targets for responsiveness	Form load – under 2 seconds; report load – under 5 seconds Approval email – within 30 seconds; task creation – within 1 minute Flow execution – under 6 seconds (observed); CMDB lookup API – under 1 second

S.No	Characteristics	Description	Technology
			Real-time UI Policies; immediate notifications; built-in caching